Percentages answer key

#	ID	Answer	Difficulty
1	026410-DC	D	Easy
2	08199-DC	B	Easy
3	374a0f38-19f3-4644-8e99-cb97adaa27f3	D	Easy
4	14922-DC	D	Easy
5	10596-DC	D	Easy
6	828689ef-29d6-4a4b-9945-060543cb1150	C	Easy
7	06469-DC	C	Easy
8	d82d4647-67d1-4ebf-8750-cf2ccee38a60	B	Easy
9	ee20c3bc-a074-467b-a5ec-dafc0c828d64	B	Easy
10	028969-DC	B	Easy
11	058dbe26-68a9-4661-8276-5e77b75ac679	D	Easy
12	baded933-3db8-4cfd-9e63-74113b894ad5	17	Easy
13	20029b3e-d1cb-42ea-9a4b-f926abe12665	B	Easy
14	029360-DC	D	Easy
15	691ec468-2184-424c-ac94-53b3d07a4270	B	Easy
16	6c847302-0dac-4944-8362-0bed796f6728	A	Easy
17	70496145-8af1-415b-9df1-cb281b9bc4d2	D	Easy
18	028060-DC	C	Easy
19	8ac6b4d3-1459-4464-8d25-73cd50d43d67	B	Easy
20	ae635e6f-331e-4b53-b528-468662017b4e	B	Easy
21	fc5659d4-90ce-4136-b70e-ac4dd697c151	D	Easy
22	d76b8468-eb1c-4309-9278-5eda6d2d9b3e	B	Easy

23	46e791de-a8dd-4add-bf7d-3a1c1fbef6ce	75	Easy
24	06907-DC	C	Easy
25	f0fb6cfb-ea46-4a28-8379-6d0acc914299	A	Easy
26	070905-DC	B	Easy
27	e06a745d-b077-48ee-8763-fdfcd713b268	A	Easy
28	71ad6fcc-0b8a-4040-a827-1cf9cc1ed910	240	Easy
29	3b2762db-3e04-4423-acad-306a8ce4ed60	161	Medium
30	0359b97d-4f74-4893-886b-a63458ac15d7	A	Medium
31	b3f57376-4374-4b2d-ab34-98d991fed767	184	Medium
32	026407-DC	A	Medium
33	b90e5782-c428-473b-8c51-a587d2c330b9	B	Medium
34	031617-DC	C	Medium
35	9d08821e-6f03-46fa-b7f1-5b1d8c3a6d3f	324	Medium
36	10b6d39c-f6c1-4d7f-a4e7-f4e5220bb807	450	Medium
37	028999-DC	C	Medium
38	db08d5bc-9768-4b87-8642-1033890325ea	C	Medium
39	f518f607-0819-47bd-b33b-db922ee03cb2	B	Medium
40	ccae50dc-b231-44c5-869e-aacfc573e043	110	Medium
41	019667-DC	C	Medium
42	05321-DC	?	Medium
43	cb21253d-b0c2-4157-bf21-8fa351eb6b79	D	Medium
44	05323-DC	D	Medium
45	c58c16c7-0525-4a20-b6ce-33c59a2d682f	C	Medium
46	967c754c-6ed6-4648-9b7b-4a17946cf25f	10	Medium
47	884dde13-7464-48c1-89f7-7a74bb1e025e	C	Medium
48	47d32f6a-d945-4d6e-9f0f-18a983b72be4	56	Medium

49	c709983b-f5d2-4f76-8755-5eb306b94baa	27	Medium
50	2ac7bdf3-7d6c-4145-a9a6-af9bf15e1765	52	Medium
51	00142-DC	B	Hard
52	65f570ee-64b0-4f20-8200-435cc6939ac8	D	Hard
53	8b00175d-6fbb-4911-bbd3-f2a1167fb3e5	B	Hard
54	9aa8fbdb-6e32-4b91-81d7-831d0518641e	B	Hard
55	a441304f-3293-46ff-a2c2-cdbb915b8b11	9.87, 987/100	Hard
56	0e0b2da3-40d6-47ae-8c52-727851d9038b	203/50, 4.06	Hard
57	ad721785-5e4c-4d71-bf8a-1acab4691b6c	D	Hard
58	22570c4d-ed8e-4040-ae8a-171921cd4f53	C	Hard
59	a643565e-80af-4950-9896-22d6389302c4	D	Hard
60	29c0769a-6466-4e2e-8793-1978ec02749c	189/4, 47.25	Hard
61	38d4bbcc-99b8-4fc3-8247-5add78d4c1dd	.88, 22/25	Hard
62	5cbbaf2c-1ba2-4545-b2ad-f26827975dbc	B	Hard
63	dabe922d-3778-48a9-88e7-926763881b51	C	Hard
64	989b4b44-3e81-4a7a-a64c-9ab381c1b983	600	Hard
65	ee0361d9-4c33-463e-a8a8-6830e04fd051	1.8, 9/5	Hard
66	5c387bc7-97b0-4bcf-b06a-892858d9b7b1	79	Hard
67	101896fa-20f1-44ca-a5ec-1b422cb3fce3	1512/5, 302.4	Hard
68	04778-DC	?	Hard
69	91c12a4d-8406-4ac0-bd8c-732595daf5a4	.54, 27/50	Hard
70	1edfed1c-89c5-4702-a448-e023544a67b4	4.05, 81/20	Hard
71	36bef1c8-d853-462f-a9e8-7575812e84e9	60000	Hard
72	fb4b074b-8096-42a7-9ab5-1672ab4556c2	A	Hard
73	f3910fb9-490b-407a-b587-704499bf653f	.0014	Hard
74	b52952eb-21e9-41d5-ac01-117586367fc5	.48, 12/25	Hard

75	af7c40ef-acf4-4f14-9563-4a4d1a99a37d	D	Hard
76	0eac6bca-d744-4e7a-be68-7d4922a300ce	2.31	Hard
77	070908-DC	?	Hard
78	070906-DC	B	Hard
79	fa595c60-481e-4838-966e-bb53dd5ec839	D	Hard
80	71e0f3ea-a756-49f9-bc0d-dbe87bc5a198	A	Hard
81	58739008-cda2-4703-9c33-924423c8a7d7	576	Hard
82	02ff4aa9-be78-434c-a64d-d18b83b4e58a	A	Hard
83	8edf9775-0d89-4ffa-8b53-27e4265b71f6	B	Hard
84	bd6a0437-de79-407d-a6f8-ad2d753ff04e	D	Hard

Rationales

▼ 1 — D

Choice D is correct. It's given that the original price of the jacket has been reduced by 20%. Multiplying the original price, $50, by 20% gives the amount, in dollars, that the price of the jacket is reduced by: $50 \times .20 = 10$. Subtracting this value from the original price results in the sale price of the jacket: $\$50 - \$10$, or $40.

Choices A, B, and C are incorrect and may result from a conceptual or calculation error.

▼ 2 — B

Choice B is correct. The percent of Employee A's calls that received a rating of 1 star is the number of Employee A's 1-star calls divided by the total number of Employee A's calls. This quotient, $\frac{16}{145}$, is approximately equal to 0.1103, or 11.03%. To the nearest whole percent, this is 11%.

Choice A is incorrect. This is the percent of all calls taken by Employee A that received a rating of 1 star. Choice C is incorrect and may result from a conceptual error. For example,

16 is the number, not the percent, of calls taken by Employee A that received a rating of 1 star. Choice D is incorrect. This is the percent of all calls that received a rating of 1 star that were taken by Employee A.

▼ 3 — D

Choice D is correct. The proportion of the beads that are silver can be written as $\frac{828{,}000}{900{,}000}$, or 0.92. Therefore, the percentage of the beads that are silver is 0.92100, or 92%.

Choice A is incorrect. This is the percentage of the beads that are not silver.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 4 — D

Choice D is correct. The total bill was $25.00. The percentage 20% is equivalent to the decimal 0.2. The tip is the product of the percentage and the total bill; therefore, $0.2 \times 25 = 5$, so the tip was $5.00.

Choices A, B, and C are incorrect and may be the result of incorrectly converting the given percentage or a calculation error.

▼ 5 — D

Choice D is correct. There are 60 minutes in one hour, so an 8-hour workday has $(60)(8) = 480$ minutes. To calculate 15% of 480, multiply 0.15 by 480: $(0.15)(480) = 72$. Therefore, Lani spent 72 minutes of her workday in meetings.

Choice A is incorrect because 1.2 is 15% of 8, which gives the time Lani spent of her workday in meetings in hours, not minutes. Choices B and C are incorrect and may be the result of computation errors.

▼ 6 — C

Choice C is correct. It's given that during an assembly, students occupied 50% of the 840 seats in the school auditorium. Therefore, the number of seats that the students occupied

during this assembly can be calculated by multiplying the number of seats in the school auditorium by $\frac{50}{100}$. Thus, the students occupied $840\frac{50}{100}$, or 420, seats during this assembly.

Choice A is incorrect. This is approximately 3%, not 50%, of 840.

Choice B is incorrect. This is approximately 6%, not 50%, of 840.

Choice D is incorrect. This is approximately 94%, not 50%, of 840.

▼ 7 — C

Choice C is correct. According to the table, 63% of survey respondents get most of their medical information from a doctor and 13% get most of their medical information from the Internet. Therefore, 76% of the 1,200 survey respondents get their information from either a doctor or the Internet, and 76% of 1,200 is 912.

Choices A, B, and D are incorrect. According to the table, 76% of survey respondents get their information from either a doctor or the Internet. Choice A is incorrect because 865 is about 72% (the percent of survey respondents who get most of their medical information from a doctor or from magazines/brochures), not 76%, of 1,200. Choice B is incorrect because 887 is about 74%, not 76%, of 1,200. Choice D is incorrect because 926 is about 77%, not 76%, of 1,200.

▼ 8 — B

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 470 can be represented as $\frac{10}{100}470$, which is equivalent to 0.10470, or 47. Therefore, 10% of 470 is 47.

Choice A is incorrect. This is 10% of 370, not 10% of 470.

Choice C is incorrect. This is 90% of 470, not 10% of 470.

Choice D is incorrect. This is $470 - 10$, not 10% of 470.

▼ 9 — B

Choice B is correct. It's given that 4% of the 350 objects in the box are balls. Therefore, the number of balls in the box can be found by calculating 4% of 350, which is equal to $350\frac{4}{100}$, or 14.

Choice A is incorrect. This is the percentage of objects in the box that are balls, not the number of balls in the box.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 10 — B

Choice B is correct. The decimal equivalent of 20% is 0.2. The number that is 20% greater than 60 is also 120% of 60. The decimal equivalent of 120% is 1.2, and $1.2(60) = 72$.

Alternate approach: 10% of 60 is 6, and 20% of 60 is double that amount, or 12. It follows that the number that is 20% greater than 60 is 12 more than 60, or $60 + 12 = 72$.

Choice A is incorrect and may result from dividing, instead of multiplying, 60 by 1.2. Choice C is incorrect because it's 25% greater than 60, rather than 20% greater than 60. Choice D is incorrect and may result from multiplying 60 by 2.2 instead of 1.2.

▼ 11 — D

Choice D is correct. The proportion of the paper clips that are size large can be written as $\frac{234,000}{300,000}$, or 0.78. Therefore, the percentage of the paper clips that are size large is 0.78100, or 78%.

Choice A is incorrect. This is the percentage of the paper clips that are not size large.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 12 — 17

The correct answer is 17. It's given that there are 170 blocks in a container, and of these blocks, 10% are green. Since 10% can be rewritten as $\frac{10}{100}$, or 0.1, the number of green blocks in the container is 0.1170, or 17.

▼ 13 — B

Choice B is correct. It's given that the number of students enrolled in an interior design program this year is 147% of last year's number, which is 200. 147% of 200 can be expressed as $\frac{147}{100}200$, or 1.47200, which is equivalent to 294. Therefore, 294 students are enrolled in the interior design program this year.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 14 — D

Choice D is correct. The decimal equivalent of 29% is 0.29. It's given that the 113,000 occupational therapy jobs in the United States in 2012 are projected to increase by 29% by 2022. Increasing 113,000 by 29% can be expressed as $(113{,}000)(1+0.29)$, or $(113{,}000)(1.29)$. Evaluating this expression yields 145,770. The closest number is 145,800 in choice D.

Choice A is incorrect and may result from increasing 113,000 by 2,900 instead of by 29%.
Choice B is incorrect and may result from increasing 113,000 by 2.9% instead of by 29%.
Choice C is incorrect and may result from increasing 113,000 by 29,000 instead of by 29%.

▼ 15 — B

Choice B is correct. The number of harvested potatoes Isabel saved to plant next year can be calculated by multiplying the total number of potatoes Isabel harvested, 760, by the proportion of potatoes she saved. Since she saved 10% of the potatoes she harvested, the proportion of potatoes she saved is $\frac{10}{100}$, or 0.1. Multiplying 760 by this proportion gives 7600.1, or 76, potatoes that she saved to plant next year.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 16 — A

Choice A is correct. It's given that 10% of the 320 marbles in a container are red. Therefore, the number of red marbles can be calculated by multiplying the number of marbles in the container by $\frac{10}{100}$, which gives $320\frac{10}{100}$, or 32.

Choice B is incorrect. This is the number of marbles in the container that aren't red.

Choice C is incorrect. This is the total number of marbles in the container.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 17 — D

Choice D is correct. It's given that the length of the base of the parallelogram is 89% of the height of the parallelogram. Since h is the height of the parallelogram, it follows that the length of the base of the parallelogram can be represented by the expression $\frac{89}{100}h$, or $0.89h$.

Choice A is incorrect. This expression represents $8{,}900\%$, not 89%, of the height of the parallelogram.

Choice B is incorrect. This expression represents 8.9%, not 89%, of the height of the parallelogram.

Choice C is incorrect. This expression represents 890%, not 89%, of the height of the parallelogram.

▼ 18 — C

Choice C is correct. The total cost to purchase the shirt is the $20 cost of the shirt plus the 5% sales tax. The value of the 5% sales tax on the $20 shirt is equivalent to $(0.05)(\$20)$, or $1. Therefore, the total cost to purchase the shirt is $\$20 + \$1$, or $21.

Choice A is incorrect and may result from neglecting to multiply by $20 when finding the value of the sales tax. Choice B is incorrect and may result from dividing by 10, instead of by 100, and then neglecting to multiply by $20 when finding the sales tax. Choice D is incorrect and may result from interpreting the sales tax of 5% as $5.

▼ 19 — B

Choice B is correct. 20% of 440 can be calculated as $\frac{20}{100}440$, which is equivalent to $\frac{8{,}800}{100}$, or 88.

Choice A is incorrect. This is 10%, not 20%, of 440.

Choice C is incorrect. This is 200%, not 20%, of 440.

Choice D is incorrect. This is 400%, not 20%, of 440.

▼ 20 — B

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 370 can be represented as $\frac{10}{100}370$, which is equivalent to 0.10370, or 37. Therefore, 10% of 370 is 37.

Choice A is incorrect. This is 10% of 270, not 10% of 370.

Choice C is incorrect. This is 90% of 370, not 10% of 370.

Choice D is incorrect. This is $370 - 10$, not 10% of 370.

▼ 21 — D

Choice D is correct. Let x represent the number that 21 is 21% of. It follows that $\frac{21}{x} = \frac{21}{100}$. Multiplying each side of this equation by x yields $21 = \frac{21x}{100}$. Multiplying each side of this equation by 100 yields $2{,}100 = 21x$. Dividing each side of this equation by 21 yields $100 = x$. Therefore, 21 is 21% of 100.

Choice A is incorrect. 21% of 0 is 0, not 21.

Choice B is incorrect. 21% of 1 is 0.21, not 21.

Choice C is incorrect. 21% of 42 is 8.82, not 21.

▼ 22 — B

Choice B is correct. It's given that there are 250 trees in a park and of these trees, 6% are birch trees. The number of birch trees in the park can be calculated by multiplying the number of trees in the park by $\frac{6}{100}$. Therefore, the number of birch trees in the park is $250\frac{6}{100}$, or 15.

Choice A is incorrect. This is the percentage of trees in the park that are birch trees, not the number of birch trees in the park.

Choice C is incorrect. This is 30%, not 6%, of 250.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 23 — 75

The correct answer is 75. It's given that Amara harvested 750 cherry tomatoes and gave 10% of them to her neighbor. 10% can be written as $\frac{10}{100}$, or 0.1. Therefore, the number of cherry tomatoes that Amara gave to her neighbor can be calculated by multiplying the total number of cherry tomatoes by 0.1, which gives $(0.1)(750)$, or 75.

▼ 24 — C

Choice C is correct. If the bill is $78.20, 15% of the bill can be found by multiplying 78.20 by the decimal conversion of 15%, 78.20 × 0.15 = $11.73. The exact amount $11.73 is closest in value to $12.00.

Choices A, B, and D are incorrect and may be the result of errors when calculating 15% of the total $78.20.

▼ 25 — A

Choice A is correct. 23% of 100 can be calculated by multiplying $\frac{23}{100}$ by 100, which yields $\frac{23}{100}100$, or 23.

Choice B is incorrect. This is 46%, not 23%, of 100.

Choice C is incorrect. This is 23% less than 100, not 23% of 100.

Choice D is incorrect. This is 23% greater than 100, not 23% of 100.

▼ 26 — B

Choice B is correct. According to the table, the minimum wage in 1960 was $1.00 per hour, and in 1970 it was $1.60 per hour. The percentage change is therefore $100\left(\frac{1.60-1.00}{1.00}\right)=60\%$.

Choice A is incorrect and may result from averaging the two wages before calculating the percentage change. Choice C is incorrect. This is the 1960 wage expressed as a percentage of the 1970 wage, not the percentage change between the two. Choice D is incorrect and may result from a calculation error.

▼ 27 — A

Choice A is correct. Let x represent the percentage of 300 that is 75. This can be written as $\frac{x}{100}300 = 75$, or $3x = 75$. Dividing both sides of this equation by 3 yields $x = 25$. Therefore, 25% of 300 is 75.

Choice B is incorrect. 50% of 300 is 150, not 75.

Choice C is incorrect. 75% of 300 is 225, not 75.

Choice D is incorrect. 225% of 300 is 675, not 75.

▼ 28 — 240

The correct answer is 240. It's given that 80% of the 300 seeds sprouted. Therefore, the number of seeds that sprouted can be calculated by multiplying the number of seeds that were planted by $\frac{80}{100}$, which gives $300\frac{80}{100}$, or 240.

▼ 29 — 161

The correct answer is 161. For a number to be 40% greater than 115, it follows that the number is $100\%\ \text{ of }\ 115 + 40\%\ \text{ of }\ 115$, which can be written as $\frac{100}{100}115 + \frac{40}{100}115$. This expression is equivalent to $1115 + 0.4115$, or 1.4115, which is equal to 161. Therefore, 161 is 40% greater than 115.

▼ 30 — A

Choice A is correct. The result of increasing a positive quantity w by $x\%$ can be represented by the expression $1 + \frac{x}{100}w$. Therefore, the result of increasing a positive quantity w by 43% can be found by substituting 43 for x in the expression $1 + \frac{x}{100}w$, which gives $1 + \frac{43}{100}w$, or $1.43w$. Thus, the expression $1.43w$ represents the result of increasing a positive quantity w by 43%.

Choice B is incorrect. This is the result of decreasing a positive quantity w by 43%.

Choice C is incorrect. This is the result of increasing a positive quantity w by $4{,}200\%$.

Choice D is incorrect. This is the result of decreasing a positive quantity w by 57%.

▼ 31 — 184

The correct answer is 184. It's given in the table that there are 375 objects in group A and 690 objects in group C. It's also given that the number of objects in group C is $p\%$ of the number of objects in group A. Therefore, 690 is $p\%$ of 375, which can be represented by

$690 = \frac{p}{100}375$, or $690 = 3.75p$. Dividing both sides of this equation by 3.75 yields $184 = p$. Therefore, the value of p is 184.

▼ 32 — A

Choice A is correct. Increasing the positive quantity x by 9% is the result of adding 9% of x to x. 9% of x can be represented algebraically as $\frac{9}{100}x$, or $0.09x$. Adding this expression to x yields $x + 0.09x$, or $1.09x$.

Choice B is incorrect. This represents 9% of x. Choice C is incorrect. This represents increasing x by 9, not by 9%. Choice D is incorrect. This represents increasing x by 0.09, not by 9%.

▼ 33 — B

Choice B is correct. It's given that in a group, 40% of the items are red. It follows that the number of red items in the group can be represented by $0.4x$, where x represents the total number of items in the group. It's also given that of all the red items in the group, 30% also have stripes. It follows that the number of items in the group that are red and have stripes can be represented by $0.30.4x$, or $0.12x$. The expression $0.12x$ represents 12% of x. Since x represents the total number of items in the group, it follows that 12% of the items in the group are red and have stripes.

Choice A is incorrect and may result from subtracting 30% from 40% rather than calculating 30% of 40%.

Choice C is incorrect and may result from adding 30% and 40% rather than calculating 30% of 40%.

Choice D is incorrect and may result from calculating the percentage that 30% is of 40% rather than calculating 30% of 40%.

▼ 34 — C

Choice C is correct. It's given that the number k is 36% greater than 50. Therefore, the value of k is the number 50 plus 36% of 50. This can be rewritten as $k = 50 + \left(\frac{36}{100}\right)(50)$.

Multiplying the terms $\left(\frac{36}{100}\right)(50)$ yields 18, so $k = 50 + 18$, or $k = 68$. It's also given that k is the product of 50 and r, which can be rewritten as $k = 50r$. Substituting 68 for k yields $68 = 50r$. Dividing both sides of this equation by 50 yields $r = 1.36$.

Choice A is incorrect. This is the percentage that k is greater than 50. Choice B is incorrect and may result from a calculation error. Choice D is incorrect. This would be the value of r if k were 36% of 50, instead of 36% greater than 50.

▼ 35 — 324

The correct answer is 324. It's given that 1,250 people attended the first information session, and that 72% of the people who attended the first information session attended the second information session. Therefore, the number of people who attended the first and second information sessions can be found by calculating 72% of 1,250, which is equal to $1,250\frac{72}{100}$, or 900. It's also given that 36% of the people who attended the first and second information sessions attended the third information session. Since 900 people attended the first and second information sessions, the number of people who attended the first, second, and third information sessions can be found by calculating 36% of 900, which is equal to $900\frac{36}{100}$, or 324. Therefore, 324 people attended all three information sessions.

▼ 36 — 450

The correct answer is 450. Let x represent the number that 432 is 96% of. This can be written as $\frac{96}{100}x = 432$, or $0.96x = 432$. Dividing both sides of this equation by 0.96 yields $x = 450$. Therefore, 432 is 96% of 450.

▼ 37 — C

Choice C is correct. The decimal equivalent of 6% is 0.06. Since increasing the number n by 6% yields the number 318, this situation can be represented by the equation $n(1 + 0.06) = 318$, or $n(1.06) = 318$. Dividing both sides of this equation by 1.06 yields $n = 300$.

Choice A is incorrect. This is the result when n is increased by 60%, not by 6%. Choice B is incorrect. This is the approximate result of decreasing 318 by 6%. Choice D is incorrect. This is the approximate result of increasing 318 by 6%.

▼ 38 — C

Choice C is correct. It's given that the population of City A increased by 7% from 2015 to 2016. Therefore, the population of City A in 2016 includes 100% of the population of City A in 2015 plus an additional 7% of the population of City A in 2015. This means that the population of City A in 2016 is 107% of the population in 2015. Thus, the population of City A in 2016 is $\frac{107}{100}$, or 1.07, times the 2015 population. Therefore, the value of k is 1.07.

Choice A is incorrect. This would be the value of k if the population in 2016 was 7% of the population in 2015.

Choice B is incorrect. This would be the value of k if the population in 2016 was 70% of the population in 2015.

Choice D is incorrect. This would be the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

▼ 39 — B

Choice B is correct. It's given that the value of z is 1.13 times 100. This can be written as $z = 1.13100$, which is equivalent to $z = 1 + 0.13(100)$, or $z = 1 + \frac{13}{100}100$. It follows that the value of z is 100% of 100 plus 13% of 100. Therefore, the value of z is 13% greater than 100.

Choice A is incorrect. This gives a value of z that is 1.113, not 1.13, times 100.

Choice C is incorrect. This gives a value of z that is 2.30, not 1.13, times 100.

Choice D is incorrect. This gives a value of z that is 3.13, not 1.13, times 100.

▼ 40 — 110

The correct answer is 110. Let x represent the total number of items in the sample. It's given that 80% of the items are faulty and that there are 88 faulty items in the sample. Therefore, 80% of x is 88. Since 80% can be rewritten as $\frac{80}{100}$, it follows that $\frac{80}{100}x = 88$. Multiplying both sides of this equation by 100 yields $80x = 8{,}800$. Dividing both sides of this equation by 80 yields $x = 110$. Therefore, there are 110 total items in the sample.

▼ 41 — C

Choice C is correct. Of the 1,300,000 units sold during the first month, 15% were returned, so $(1{,}300{,}000)(0.15) = 195{,}000$ units were returned during the first month. If the units were sold and returned at the same rate for the next 5 months, then a total of $(195{,}000)(6) = 1{,}170{,}000$ smartphone units were returned during the 6-month period.

Choice A is incorrect. This is the number of units that were returned in 1 month. Choice B is incorrect. This is the number of units that were returned in 5 months. Choice D is incorrect. This is the number of units sold and not returned during the first 6 months.

▼ 42 — ?

Choice B is correct. If the stove's value depreciates by 20% of the previous year's estimated value, then each year it retains 100% – 20% = 80%, or 0.80, of the previous year's estimated value. Since the stove's value was $800 when Thomas installed it, the estimated value after two years would be (0.80)(0.80)($800) = $512.

Choice A is incorrect. This is the value of the stove if each year it had depreciated by 20% of the original value rather than by 20% of the previous year's estimated value. Choice C is incorrect and may be the result of a computational error. Choice D is incorrect. This is the estimated value of the stove 1 year after Thomas installed it, not 2 years.

▼ 43 — D

Choice D is correct. It's given that the number of coins in the collection increased from 9 to 90. It follows that the number of coins in the collection increased by $90 - 9$, or 81. Let x% represent the percentage that 81 is of 9. The value of x can be found using the proportion $\frac{81}{9} = \frac{x}{100}$, or $9 = \frac{x}{100}$. Multiplying both sides of this equation by 100 yields $900 = x$. Thus, when the number of coins in the collection increased from 9 to 90, the percent increase was 900%.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 44 — D

Choice D is correct. To find the percent increase of the customer's water bill, the absolute increase of the bill, in dollars, is divided by the original amount of the bill, and the result is

multiplied by 100%, as follows: $\frac{79.86-75.74}{75.74} \approx 0.054$; $0.054 \times 100\% = 5.4\%$.

Choice A is incorrect. This choice is the difference $79.86-75.74$ rounded to the nearest tenth, which is the (absolute) increase of the bill's amount, not its percent increase. Choice B is incorrect and may be the result of some calculation errors. Choice C is incorrect and is the result of dividing the difference between the two bill amounts by the new bill amount instead of the original bill amount.

▼ 45 — C

Choice C is correct. Let x be the 2015 population of Greenville. It's given that the population increased by 7% from 2015 to 2016. The increase in population can be written as $0.07x$. The 2016 population of Greenville is given as the sum of the 2015 population of Greenville and the increase in population from 2015 to 2016. This can be rewritten as $x + 0.07x$, or $1.07x$. Therefore, the value of k is 1.07.

Choice A is incorrect. This is the percent, represented as a decimal, that the population increased from 2015 to 2016, not the value of k.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

▼ 46 — 10

The correct answer is 10. It's given that the amount of Hanna's food order was $\$ 50$ and that Hanna gave a tip of 20% of the amount of the bill. 20% of 50 can be calculated as $\frac{20}{100}50$, which yields $\frac{1000}{100}$, or 10. Therefore, the amount, in dollars, of the tip Hanna gave is 10.

▼ 47 — C

Choice C is correct. It's given that the price of the collectible card was $\$ 15.50$ in March and $\$ 17.36$ in April. It's also given that the price of the collectible card in April was $p\%$ of the price in March. It follows that $\$ 17.36$ is $p\%$ of $\$ 15.50$. Therefore, the value of p can be calculated by solving the equation $17.36 = \frac{p}{100}15.50$, or $17.36 = \frac{15.50p}{100}$. Multiplying each side of this equation by 100 yields $1{,}736 = 15.50p$. Dividing each side of this equation by 15.50 yields $112 = p$. Therefore, the value of p is 112.

Choice A is incorrect. 12% is the percent increase in the price of the collectible card from March to April.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 48 — 56

The correct answer is 56. It's given that Cedric had 35 plants in his garden last year and that the number of plants in Cedric's garden this year is 60% greater than the number of plants in his garden last year. It follows that the number of plants in Cedric's garden this year is 35 plus 60% of 35, which is equal to $35 + 35\frac{60}{100}$, or $35 + 350.6$. This expression is equivalent to $35 + 21$, or 56. Therefore, Cedric has 56 plants in his garden this year.

▼ 49 — 27

The correct answer is 27. It's given that 6% of the 450 tiles in a box are black. Therefore, the number of black tiles in the box can be calculated by multiplying the number of tiles in the box by $\frac{6}{100}$, which is equivalent to $450\frac{6}{100}$, or 27.

▼ 50 — 52

The correct answer is 52. It's given that 13 is p% of 25. It follows that $\frac{13}{25} = \frac{p}{100}$. Multiplying both sides of this equation by 100 gives $52 = p$. Therefore, the value of p is 52.

▼ 51 — B

Choice B is correct. The percent increase from 2012 to 2013 was $\frac{5{,}880 - 5{,}600}{5{,}600} = 0.05$, or 5%. Since the percent increase from 2012 to 2013 was estimated to be double the percent increase from 2013 to 2014, the percent increase from 2013 to 2014 was expected to be 2.5%.
Therefore, the number of subscriptions sold in 2014 is expected to be the number of subscriptions sold in 2013 multiplied by $(1 + 0.025)$, or $5{,}880(1.025) = 6{,}027$.

Choice A is incorrect and is the result of adding half of the value of the increase from 2012 to 2013 to the 2013 result. Choice C is incorrect and is the result adding twice the value of the increase from 2012 to 2013 to the 2013 result. Choice D is incorrect and is the result of interpreting the percent increase from 2013 to 2014 as double the percent increase from 2012 to 2013.

▼ 52 — D

Choice D is correct. Let $n\%$ represent the percent by which the positive quantity x is decreased to result in $0.35x$. The value of n can be found by solving the equation $x - \frac{n}{100}x = 0.35x$. Since x is a common factor of each of the terms on the left-hand side of this equation, the equation can be rewritten as $x1 - \frac{n}{100} = 0.35x$. Dividing each side of this equation by x yields $1 - \frac{n}{100} = 0.35$. Multiplying each side of this equation by 100 yields $100 - n = 35$. Subtracting 100 from each side of this equation yields $-n = -65$. Dividing each side of this equation by -1 yields $n = 65$. Therefore, the expression $0.35x$ represents the result of decreasing the positive quantity x by 65%.

Choice A is incorrect. Decreasing the quantity x by 3.5% yields $x - 0.035x$, or $0.965x$, not $0.35x$.

Choice B is incorrect. Decreasing the quantity x by 35% yields $x - 0.35x$, or $0.65x$, not $0.35x$.

Choice C is incorrect. Decreasing the quantity x by 6.5% yields $x - 0.065x$, or $0.935x$, not $0.35x$.

▼ 53 — B

Choice B is correct. It's given that the number b is 80% less than 24. It follows that b is equal to 24 minus 80% of 24, which can be written as $b = 24 - \frac{80}{100}24$. This is equivalent to $b = 24 - 0.824$, or $b = 4.8$. It's also given that the number a is 190% greater than the number b. It follows that a is equal to b plus 190% of b. Since $b = 4.8$, this can be written as $a = 4.8 + \frac{190}{100}4.8$. This is equivalent to $a = 4.8 + 1.94.8$, or $a = 13.92$.

Choice A is incorrect. This would be the value of a if a were 190% of b, not 190% greater than b.

Choice C is incorrect. This is $190 - 80\%$ of 24.

Choice D is incorrect. This would be the value of a if b were 80% of 24, not 80% less than 24, and a were 190% of b, not 190% greater than b.

▼ 54 — B

Choice B is correct. It's given that 20% of the original number of marbles were removed from the group. Let x represent the original number of marbles in the group. It follows that $\frac{20}{100}x$, or $0.20x$, marbles were removed from the group. Therefore, $x - 0.20x$ marbles remained in the group. It's also given that 360 marbles remained in the group. Thus, $x - 0.20x = 360$, or

$0.80x = 360$. Dividing both sides of this equation by 0.80 yields $x = 450$. Substituting 450 for x in the expression $0.20x$ yields $0.20(450)$, or 90. Therefore, 90 marbles were removed from the group.

Choice A is incorrect. This is 20% of the remaining number of marbles.

Choice C is incorrect. This is the original number of marbles, not the number of marbles that were removed.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 55 — 9.87, 987/100

The correct answer is 9.87. It's given that the number a is 110% greater than the number b. It follows that $a = 1 + \frac{110}{100}b$, or $a = 2.1b$. It's also given that the number b is 90% less than 47. It follows that $b = 1 - \frac{90}{100}47$, or $b = 0.147$, which yields $b = 4.7$. Substituting 4.7 for b in the equation $a = 2.1b$ yields $a = 2.14.7$, which is equivalent to $a = 9.87$. Therefore, the value of a is 9.87.

▼ 56 — 203/50, 4.06

The correct answer is 4.06. It's given that the retail price is 290% of the wholesale price of $\$ 7.00$. Thus, the retail price is $\$ 7.00\frac{290}{100}$, which is equivalent to $\$ 7.002.9$, or $\$ 20.30$. It's also given that the discounted price is 80% off the retail price. Thus, the discounted price is $\$ 20.301 - \frac{80}{100}$, which is equivalent to $\$ 20.300.20$, or $\$ 4.06$.

▼ 57 — D

Choice D is correct. It's given that a is $2{,}241\%$ of the sum of b and c. This can be represented by the equation $a = \frac{2{,}241}{100}b + c$, or $a = 22.41b + c$. It's also given that that b is 83% of c. This can be represented by the equation $b = \frac{83}{100}c$, or $b = 0.83c$. Dividing both sides of this equation by 0.83 yields $\frac{b}{0.83} = c$. Substituting $\frac{b}{0.83}$ for c in the equation $a = 22.41b + c$ yields $a = 22.41b + \frac{b}{0.83}$, or $a = 22.41\frac{1.83b}{0.83}$, which is equivalent to $a = \frac{41.0103b}{0.83}$, or $a = 49.41b$. This equation is equivalent to $a = \frac{4{,}941}{100}b$; therefore, a is $4{,}941\%$ of b.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 58 — C

Choice C is correct. It's given that the total number of hours visitors used the exercise room increased from t hours in July to $3.49t$ hours in August. It follows that the number of hours increased by $3.49t - t$, or $2.49t$. Let $p\%$ represent the percent increase in the exercise room use from July to August, or the percentage that $2.49t$ is of t. The value of p can be found using the proportion $\frac{2.49t}{t} = \frac{p}{100}$, or $2.49 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p = 249$. Therefore, the percent increase in the number of hours visitors used the exercise room from July to August is 249%.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. The exercise room use in August is 349% of the exercise room use in July, but this doesn't represent the percent increase from July to August.

▼ 59 — D

Choice D is correct. It's given that the speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is $5{,}900\%$ of the speed of object C, this means that $a = \frac{5{,}900}{100}c$, or $a = 59c$. If the speed of object C is 0.008% of the speed of object B, this means that $c = \frac{0.008}{100}b$, or $c = 0.00008b$. Dividing both sides of this equation by 0.00008 yields $\frac{1}{0.00008}c = b$, or $12{,}500c = b$. Since $a = 59c$ and $b = 12{,}500c$, it follows that $a + b = 59c + 12{,}500c$, or $a + b = 12{,}559c$. Therefore, the expression $12{,}559c$ represents the value of $a + b$ in terms of c.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 60 — 189/4, 47.25

The correct answer is 47.25. It's given that the number w is 110% greater than the number z. It follows that $w = 1 + \frac{110}{100}z$, or $w = 2.1z$. It's also given that the number z is 55% less than 50. It follows that $z = 1 - \frac{55}{100}50$, or $z = 0.4550$, which yields $z = 22.5$. Substituting 22.5 for z in the equation $w = 2.1z$ yields $w = 2.122.5$, which is equivalent to $w = 47.25$. Therefore, the value of w is 47.25. Note that 47.25 and 189/4 are examples of ways to enter a correct answer.

▼ 61 — .88, 22/25

The correct answer is .88 . It's given that the number a is 60% greater than the positive number b. Therefore, $a = 1 + \frac{60}{100}b$, which is equivalent to $a = 1 + 0.60b$, or $a = 1.60b$. It's also given that the number c is 45% less than a. Therefore, $c = 1 - \frac{45}{100}a$, which is equivalent to $c = 1 - 0.45a$, or $c = 0.55a$. Since $a = 1.60b$, substituting $1.60b$ for a in the equation $c = 0.55a$ yields $c = 0.551.60b$, or $c = 0.88b$. Thus, the number c is 0.88 times the number b. Note that .88 and 22/25 are examples of ways to enter a correct answer.

▼ 62 — B

Choice B is correct. It's given that 1 inch on the drawing represents an actual distance of 19 miles and that the larger drawing has a side length that is 65% longer than the side length of the previous drawing. It follows that $(1.65)(1 \text{ inch})$, or 1.65 inches, on the larger drawing also represents an actual distance of 19 miles. Therefore, the scale of the larger drawing can be represented as $\frac{19 \text{ miles}}{1.65 \text{ inches}}$, which is approximately 11.52 miles per inch. Therefore, 11.52 is closest to the actual distance, in miles, represented by 1 inch on the larger drawing.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 63 — C

Choice C is correct. It's given that 37% of the items in a box are green. Let x represent the total number of items in the box. It follows that $\frac{37}{100}x$, or $0.37x$, items in the box are green. It's also given that of those, 37% are also rectangular. Therefore, $\frac{37}{100}0.37x$, or $0.1369x$, items in the box are green rectangular items. It's also given that of the green rectangular items, 42% are also metal. Therefore, $\frac{42}{100}0.1369x$, or $0.057498x$, items in the box are rectangular green metal items. The number of the items in the box that are not rectangular green metal items is the total number of items in the box minus the number of rectangular green metal items in the box. Therefore, the number of items in the box that are not rectangular green metal items is $x - 0.057498x$, or $0.942502x$. The percentage of items in the box that are not rectangular green metal items is the percentage that $0.942502x$ is of x. If $p\%$ represents this percentage, the value of p is $100\frac{0.942502x}{x}$, or 94.2502. Of the given choices, 94.25% is closest to the percentage of items in the box that are not rectangular green metal items.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 64 — 600

The correct answer is 600. It's given that 210 is p% greater than 30. It follows that $210 = 1 + \frac{p}{100}30$. Dividing both sides of this equation by 30 yields $7 = 1 + \frac{p}{100}$. Subtracting 1 from both sides of this equation yields $6 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p = 600$. Therefore, the value of p is 600.

▼ 65 — 1.8, 9/5

The correct answer is 1.8. It's given that the regular price of a shirt at a store is $\$ 11.70$, and the sale price of the shirt is 80% less than the regular price. It follows that the sale price of the shirt is $\$ 11.701 - \frac{80}{100}$, or $\$ 11.701 - 0.8$, which is equivalent to $\$ 2.34$. It's also given that the sale price of the shirt is 30% greater than the store's cost for the shirt. Let x represent the store's cost for the shirt. It follows that $2.34 = 1 + \frac{30}{100}x$, or $2.34 = 1.3x$. Dividing both sides of this equation by 1.3 yields $x = 1.80$. Therefore, the store's cost, in dollars, for the shirt is 1.80. Note that 1.8 and 9/5 are examples of ways to enter a correct answer.

▼ 66 — 79

The correct answer is 79. Let x represent the number of zebras in the population in 2014 and let y represent the number of zebras in the population in 2018. It's given that the number of zebras in this population in 2018 was 1.27 times the number of zebras in this population in 2014. It follows that the equation $y = 1.27x$ represents this situation. Dividing both sides of this equation by 1.27 yields $\frac{y}{1.27} = x$, or $\frac{1}{1.27}y = x$. Therefore, the number of zebras in this population in 2014 is $\frac{1}{1.27}$ times the number of zebras in this population in 2018. If the number of zebras in this population in 2014 is p% of the number of zebras in this population in 2018, then $x = \frac{p}{100}y$. It follows that $\frac{1}{1.27} = \frac{p}{100}$, or $\frac{100}{1.27} = p$, which means p is approximately equal to 78.74. Therefore, the value of p, to the nearest whole number, is 79.

▼ 67 — 1512/5, 302.4

The correct answer is 302.4. It's given that the number b is 320% greater than 160. It follows that b is equal to 160 plus 320% of 160, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is 55% less than the number b. It follows that a is equal to b minus 55% of b, which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting 672 for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.

▼ 68 — ?

The correct answer is 8. The 6 students represent $(100-15-45-25)\% = 15\%$ of those invited to join the committee. If x people were invited to join the committee, then $0.15x = 6$. Thus, there were $\frac{6}{0.15} = 40$ people invited to join the committee. It follows that there were $0.45(40) = 18$ teachers and $0.25(40) = 10$ school and district administrators invited to join the committee. Therefore, there were 8 more teachers than school and district administrators invited to join the committee.

▼ 69 — .54, 27/50

The correct answer is .54 . It's given that the number a is 70% less than the positive number b. Therefore, $a = 1 - \frac{70}{100}b$, which is equivalent to $a = 1 - 0.70b$, or $a = 0.30b$. It's also given that the number c is 80% greater than a. Therefore, $c = 1 + \frac{80}{100}a$, which is equivalent to $c = 1 + 0.80a$, or $c = 1.80a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.80a$ yields $c = 1.800.30b$, or $c = 0.54b$. Thus, c is 0.54 times b. Note that .54 and 27/50 are examples of ways to enter a correct answer.

▼ 70 — 4.05, 81/20

The correct answer is 4.05. It's given that the purchase price for certain books is 5.00 dollars each. It's also given that each book is marked for sale at a consumer price that is 270% of the purchase price. Since the consumer price is 270% of the purchase price of 5.00 dollars, it follows that the consumer price is $(2.7)(5.00)$, or 13.50, dollars. It's given that after 4 months, any remaining books are discounted at 70% off the consumer price. Thus, the discount amount is $(0.7)(13.50)$, or 9.45, dollars. Subtracting the discount amount from the consumer price gives the discounted price of each of the remaining books: $13.50 - 9.45 = 4.05$. Note that 4.05 and 81/20 are examples of ways to enter a correct answer.

▼ 71 — 60000

The correct answer is $60{,}000$. It's given that the weight of sample A is 468% of the weight of sample B and 0.780% of the weight of sample C. Let the weight of sample B be b and the weight of sample C be c. Then, the weight of sample A is equivalent to $\frac{468b}{100}$, or $4.68b$, and also equivalent to $\frac{0.780c}{100}$, or $0.00780c$. Setting these expressions equal to each other yields $4.68b = 0.00780c$. Dividing both sides of this equation by 0.00780 yields $\frac{4.68b}{0.00780} = c$, or $c = 600b$. If the weight of sample C is $p\%$ of the weight of sample B, then $c = \frac{p}{100}b$. Since $c = 600b$

and $c = \frac{p}{100}b$, it follows that $600b = \frac{p}{100}b$, or $600 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $60{,}000 = p$. Therefore, the value of p is $60{,}000$.

▼ 72 — A

Choice A is correct. It's given that the result of increasing the quantity x by 400% is 60. This can be written as $x + \frac{400}{100}x = 60$, which is equivalent to $x + 4x = 60$, or $5x = 60$. Dividing each side of this equation by 5 yields $x = 12$. Therefore, the value of x is 12.

Choice B is incorrect. The result of increasing the quantity 15 by 400% is 75, not 60.

Choice C is incorrect. The result of increasing the quantity 240 by 400% is $1{,}200$, not 60.

Choice D is incorrect. The result of increasing the quantity 340 by 400% is $1{,}700$, not 60.

▼ 73 — .0014

The correct answer is .0014 . It's given that a certain type of substance can contain a maximum of 0.001% phosphorus by mass to meet a set of standards. If a sample of the substance has a mass of 140 grams, it follows that the maximum mass, in grams, of phosphorus the sample can contain to meet the standards is 0.001% of 140, or $\frac{0.001}{100}140$, which is equivalent to 0.00001140, or 0.0014. Note that .0014 and 0.001 are examples of ways to enter a correct answer.

▼ 74 — .48, 12/25

The correct answer is .48 . It's given that the number a is 70% less than the positive number b. Therefore, $a = 1 - \frac{70}{100}b$, which is equivalent to $a = 1 - 0.70b$, or $a = 0.30b$. It's also given that the number c is 60% greater than a. Therefore, $c = 1 + \frac{60}{100}a$, which is equivalent to $c = 1 + 0.60a$, or $c = 1.60a$. Since $a = 0.30b$, substituting $0.30b$ for a in the equation $c = 1.60a$ yields $c = 1.600.30b$, or $c = 0.48b$. Thus, c is 0.48 times b. Note that .48 and 12/25 are examples of ways to enter a correct answer.

▼ 75 — D

Choice D is correct. It's given that the result of increasing the quantity x by $1{,}800\%$ is 684. It follows that $x + \frac{1{,}800}{100}x = 684$, which is equivalent to $x + 18x = 684$, or $19x = 684$. Dividing each side of this equation by 19 yields $x = 36$. Therefore, the value of x is 36.

Choice A is incorrect. The result of increasing the quantity $12{,}996$ by $1{,}800\%$ is $246{,}924$, not 684.

Choice B is incorrect. The result of increasing the quantity $12{,}312$ by $1{,}800\%$ is $233{,}928$, not 684.

Choice C is incorrect. The result of increasing the quantity 38 by $1{,}800\%$ is 722, not 684.

▼ 76 — 2.31

The correct answer is 2.31. It's given that the number of audiobooks in the library increased by 131% from last year to this year. Converting 131% to a decimal yields 1.31. It's also given that the number of audiobooks in the library last year was n, where n is a constant. It follows that the number of audiobooks in the library this year can be represented by n plus 131% of n, or $n + 1.31n$. Combining like terms in this expression yields $2.31n$. Since the number of audiobooks in the library this year is represented as bn, where b is a constant, it follows that the value of b is 2.31.

▼ 77 — ?

The correct answer is 75. The percent decrease between two values is found by dividing the difference between the two values by the original value and multiplying by 100. The line graph shows that the annual snowfall in 2003 was 40 inches, and the annual snowfall in 2007 was 10 inches. Therefore, the percent decrease in the annual snowfall from 2003 to 2007 is $\left(\frac{40-10}{40}\right)(100)$, or 75. It's given that this is equivalent to $p\%$, so the value of p is 75.

▼ 78 — B

Choice B is correct. It's given that each serving of Crunchy Grain cereal provides 5% of an adult's daily allowance of potassium, so x servings would provide x times 5%. The percentage of an adult's daily allowance of potassium, p, is 5 times the number of servings, x. Therefore, the percentage of an adult's daily allowance of potassium can be expressed as $p = 5x$.

Choices A, C, and D are incorrect and may result from incorrectly converting 5% to its decimal equivalent, which isn't necessary since p is expressed as a percentage. Additionally, choices C and D are incorrect because the context should be represented by a linear relationship, not by an exponential relationship.

▼ 79 — D

Choice D is correct. It's given that a is 230% of b. It follows that $a = \frac{230}{100}b$. It's also given that a is 60% of c. It follows that $a = \frac{60}{100}c$. Since $a = \frac{230}{100}b$ and $a = \frac{60}{100}c$, it follows that $\frac{230}{100}b = \frac{60}{100}c$. Multiplying each side of this equation by $\frac{100}{60}$ yields $\frac{23}{6}b = c$. If c is $p\%$ of b, then $c = \frac{p}{100}b$. It follows that $\frac{23}{6} = \frac{p}{100}$. Multiplying each side of this equation by 100 yields $\frac{2,300}{6} = p$. It follows that the value of p is approximately 383.33. Therefore, of the given choices, 383 is closest to the value of p.

Choice A is incorrect. This is closest to the value of p if b is 230% of a, rather than if a is 230% of b, and if b is $p\%$ of c, rather than if c is $p\%$ of b.

Choice B is incorrect. This is closest to the value of p if a is 230% greater than b, rather than 230% of b.

Choice C is incorrect. This is closest to the value of p if c is $p\%$ greater than b, rather than $p\%$ of b.

▼ 80 — A

Choice A is correct. It’s given that a scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. It’s also given that on February 15, there were 99 dragonflies in the habitat and that the percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. This can be represented by the equation
$99 = 1 + \frac{12.50}{100}x$, where x represents the number of dragonflies in the habitat on January 1. This equation can be rewritten as $99 = 1.125x$. Dividing both sides of this equation by 1.125 yields $88 = x$. Therefore, there were 88 dragonflies in the habitat on January 1.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 81 — 576

The correct answer is 576. It’s given that 0.1152 is 0.08% of the number b. It follows that $0.1152 = \frac{0.08}{100}b$, or $0.1152 = 0.0008b$. Dividing both sides of this equation by 0.0008 yields $144 = b$. It’s also given that the number b is 25% of the number c. It follows that $b = \frac{25}{100}c$, or $b = 0.25c$. Substituting 144 for b in this equation yields $144 = 0.25c$. Dividing both sides of

this equation by 0.25 yields $576 = c$. Therefore, the value of c is 576.

▼ 82 — A

Choice A is correct. It's given that the value of the comic book increased by 167% from the end of 2011 to the end of 2012. Therefore, if the value of the comic book at the end of 2011 was x dollars, then the value, in dollars, of the comic book at the end of 2012 was $x + \frac{167}{100}x$, which can be rewritten as $1x + 1.67x$, or $2.67x$. It's also given that the value of the comic book decreased by 16% from the end of 2012 to the end of 2013. Therefore, the value, in dollars, of the comic book at the end of 2013 was $2.67x - 2.67x\frac{16}{100}$, which can be rewritten as $2.67x - 2.67x0.16$, or $2.2428x$. Thus, if the value of the comic book at the end of 2011 was x dollars, and the value of the comic book at the end of 2013 was $2.2428x$ dollars, then from the end of 2011 to the end of 2013, the value of the comic book increased by $2.2428x - 1x$, or $1.2428x$, dollars. Therefore, the increase in the value of the comic book from the end of 2011 to the end of 2013 is equal to 1.2428 times the value of the comic book at the end of 2011. It follows that from the end of 2011 to the end of 2013, the net percentage increase in the value of the comic book was 1.2428100%, or 124.28%.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the difference between the net percentage increase in the value of the comic book from the end of 2011 to the end of 2012 and the net percentage decrease in the value of the comic book from the end of 2012 to the end of 2013, not the net percentage increase in the value of the comic book from the end of 2011 to the end of 2013.

Choice D is incorrect. This is the net percentage increase in the value of the comic book from the end of 2011 to the end of 2013, if the value of the comic book increased by 167% from the end of 2011 to the end of 2012 and then increased, not decreased, by 16% from the end of 2012 to the end of 2013.

▼ 83 — B

Choice B is correct. It's given that 140 is $p\%$ greater than 10. It follows that $140 = 10 + \frac{p}{100}10$, which is equivalent to $140 = 10 + \frac{10}{100}p$, or $140 = 10 + 0.1p$. Subtracting 10 from each side of this equation yields $130 = 0.1p$. Dividing each side of this equation by 0.1 yields $1{,}300 = p$, or $p = 1{,}300$.

Choice A is incorrect. This would be the value of p if 140 were $p\%$ of 10, not $p\%$ greater than 10.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 84 — D

Choice D is correct. It's given that in 2008 Zinah earned 14% more than in 2007. Let h represent the amount Zinah earned in 2007 and let j represent the amount that Zinah earned in 2008. This situation can be represented by the equation $j = 1 + \frac{14}{100}h$, or $j = 1.14h$. It's also given that in 2009 Zinah earned 4% more than in 2008. Let k represent the amount Zinah earned in 2009. This situation can be represented by the equation $k = 1 + \frac{4}{100}j$, or $k = 1.04j$. Substituting $1.14h$ for j in the equation $k = 1.04j$ yields $k = 1.041.14h$, or $k = 1.1856h$. If Zinah earned y times as much in 2009 as in 2007, then the value of y is 1.1856.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.